

DATATHON 2025

**Predicting Statistical Arbitrage Opportunities through
Market Clustering and Sequential Neural Networks**

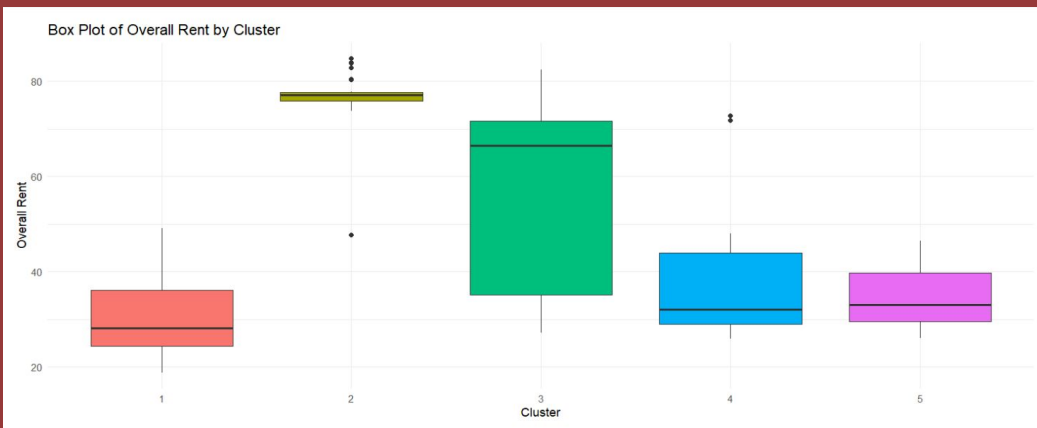
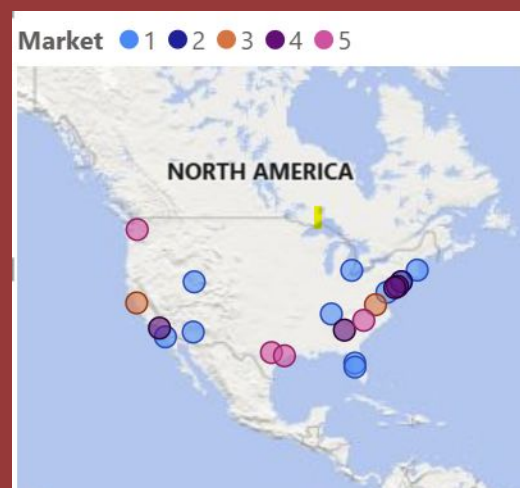
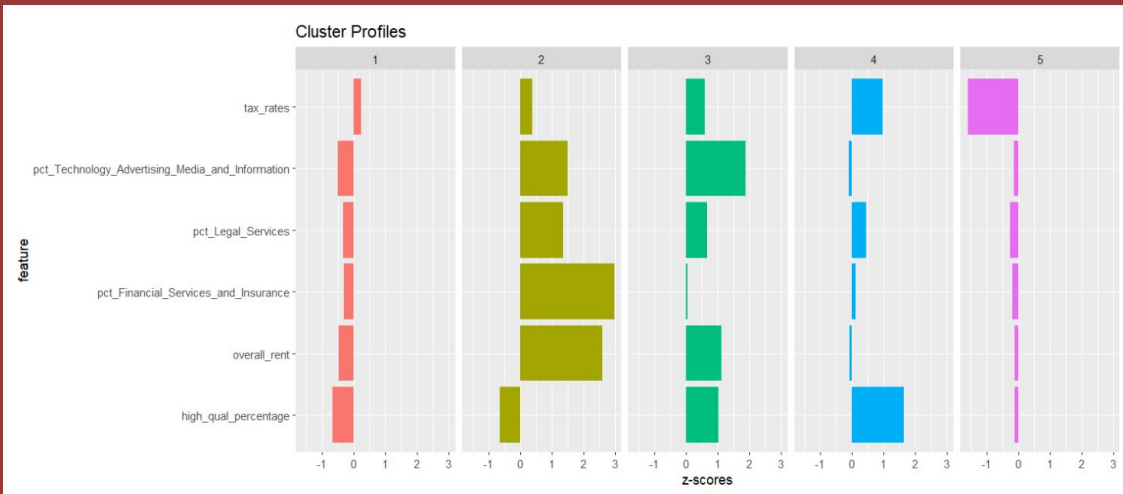
Andrusha Parilov Harden, Bonita Davis, Christopher Cook,
Davis Lynn, Sheridan McAtee

Efficient Market Hypothesis (Eugene Fama, 1970):

- Markets fully reflect all information both public and private.
- Assets are priced at a level that factors in all information.
- If markets do not perfectly reflect information, arbitrage opportunities exist for information savvy investors to make money.

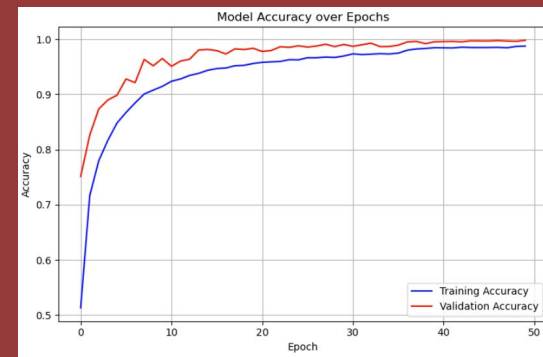
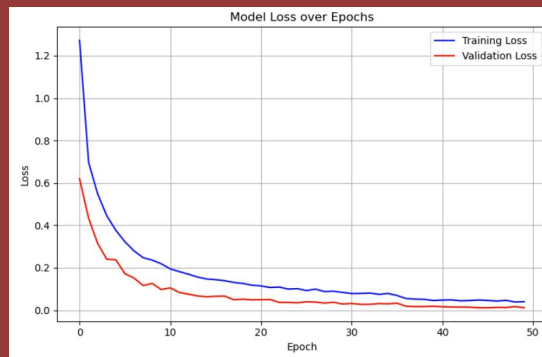
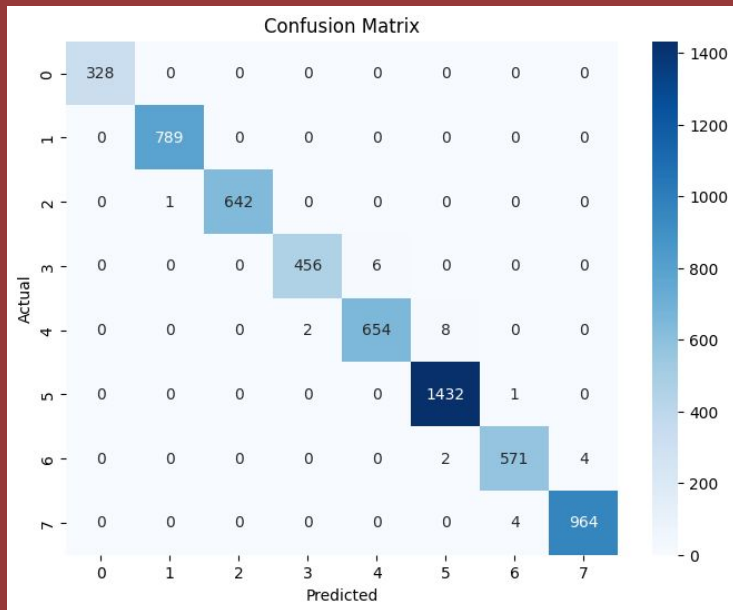
Connection to Case:

- Savills is looking for good deals for clients that match certain targeted criteria.
- Commercialized Real Estate Market lacks nearly all centralized information, and most information is fully private.
- If Savills can identify markets that are underpriced relative to the clusters they exist in, and become able to project future occupancy and rent, **Savills can effectively partake in arbitrage on behalf of their clients.**



By in large, central headquarters moves are determined by state corporate tax (Hossain & Mitra, 2023), location to competitors (Ferras-Hernandez & Nylund, 2018), stimulus from states, rent and high quality offices.

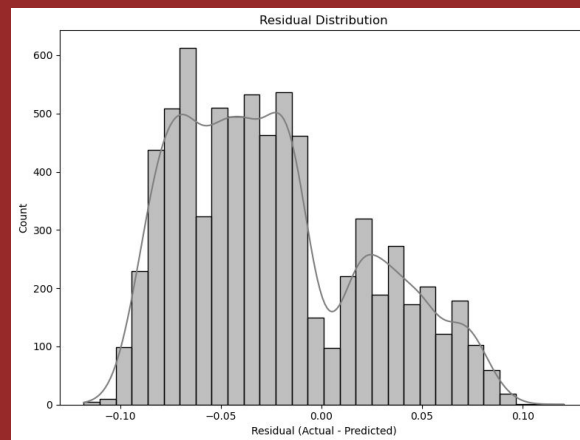
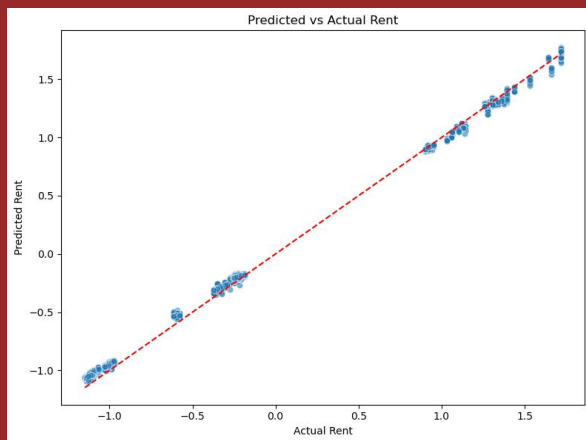
Hossain, M., & Mitra, S. (2023). Does headquarters location matter in corporate tax avoidance? *Journal of Financial Stability*, 64, 101106. <https://doi.org/10.1016/j.jfs.2023.101106>
 Ferras-Hernandez, X., & Nylund, P. A. (2018). Clusters as Innovation Engines: The Accelerating Strengths of Proximity. *European Management Review*, 16(1), 37–53. <https://doi.org/10.1111/emre.12330>



Predicting Occupancy in Octiles through Sequential Classification Neural Network:

- 3 Layers of Convolution, Pooling and Dropout
- Global Pool Layer
- Dense Layer

98% accurate in training, 100% accurate within testing dataset



Predicting Rent Prices through Regression:

- 3 layers of linear transformation, ReLU, batch transformation and dropout
- AdamW to optimize
- Early Stopping

Stopped after 8 epochs

Test MSE .0027

Test R^2 .9972

Through this project, Savills will now be able to find optimal levels of pricing and occupation within any housing market in the US.

This helps the clients find routinely underpriced markets and locations in the US, which allows them to get a better price for the level of quality they are receiving.

Furthermore, Savills will be able to provide insight into what their clients want, and place them in similar markets that may be underpriced relative to the qualities they want (i.e. Taxes, Building Quality, Location to Competitors, etc.) .